

● HARD

● ALGEBRA

Algebra

10 questions · ~15 min

07

Q1

ALGEBRA

Line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. Line j is perpendicular to line h in the xy -plane. What is the slope of line j ?

A. $-\frac{7}{5}$

B. $-\frac{5}{7}$

C. $\frac{7}{5}$

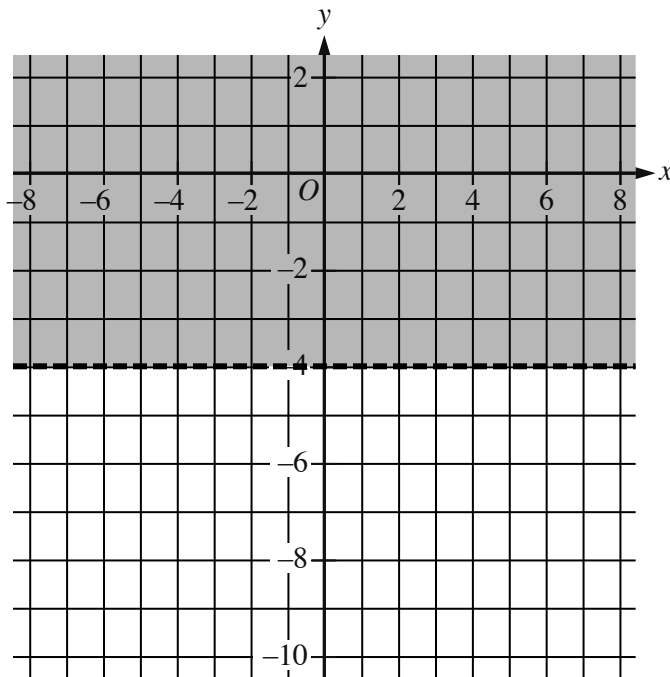
D. $\frac{5}{7}$

For a particular car, the linear function f gives the predicted power, in brake horsepower bhp, for engine speeds between 1,000 revolutions per minute rpm and 6,000 rpm. According to this function, the car's predicted power is 433 bhp at an engine speed of 3,331 rpm and 600 bhp at an engine speed of 4,500 rpm. The equation $f(x) = \frac{1}{7}x - a + 433$ defines f , where x is the engine speed, in rpm, and a is a constant. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The boundary of the inequality is a dashed line.
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 3 comma negative 4)
 - (0 comma negative 4)
 - (3 comma negative 4)
- The area above the boundary is shaded.

The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r ?

- A. 15
- B. 4
- C. -4

D. -15

Q4

ALGEBRA

$$-3x + 21px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

A. 0

B. $\frac{1}{7}$

C. $\frac{4}{3}$

D. 4

Q5

GRID-IN ALGEBRA

$$\begin{aligned} -x + y &= -3.5 \\ x + 3y &= 9.5 \end{aligned}$$

If (x, y) satisfies the system of equations above, what is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN

ALGEBRA

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $k - 5$, where k and b are constants. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

ALGEBRA

The cost of renting a carpet cleaner is \$ 52 for the first day and \$ 26 for each additional day. Which of the following functions gives the cost Cd , in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

A. $C(d) = 26d + 26$

B. $C(d) = 26d + 52$

C. $C(d) = 52d - 26$

D. $C(d) = 52d + 78$

Q8

ALGEBRA

$$y \leq x + 7$$

$$y \geq -2x - 1$$

Which point x, y is a solution to the given system of inequalities in the xy -plane?

A. $-14, 0$

B. $0, -14$

C. $0, 14$

D. $14, 0$

Q9

ALGEBRA

Each side of a 30-sided polygon has one of three lengths. The number of sides with length 8 centimeters cm is 5 times the number of sides n with length 3 cm. There are 6 sides with length 4 cm. Which equation must be true for the value of n ?

- A. $5n + 6 = 30$
- B. $6n + 6 = 30$
- C. $8n + 3n + 4n = 30$
- D. $85n + 3n + 46 = 30$

Q10

ALGEBRA

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $r, -\frac{6r}{7} + \frac{5}{7}$
- B. $r, \frac{7r}{6} + \frac{5}{6}$
- C. $\frac{r}{4} + 5, -\frac{r}{4} + 20$
- D. $-\frac{6r}{7} + \frac{5}{7}, r$